EX. NO: 3 IMPLEMENTATION OF TRIPLE DATA ENCRYPTION STANDARD (3DES)

AIM

To implement the Triple Data Encryption Standard (3DES) encryption and decryption algorithm using Java.

DESCRIPTION

Triple Data Encryption Standard (3DES) is an enhanced version of the DES algorithm that applies the DES encryption process three times using either two or three different secret keys. It provides much stronger security than DES by increasing the effective key length. The most common operation is Encrypt–Decrypt–Encrypt (EDE). Triple DES is widely used in banking systems, secure communications, and financial transactions.

ALGORITHM

STEP-1: Read the plaintext message from the user.

STEP-2: Generate a Triple DES (DESede) secret key.

STEP-3: Initialize the Triple DES cipher in encryption mode.

STEP-4: Encrypt the plaintext using the secret key.

STEP-5: Display the encrypted ciphertext.

STEP-6: Initialize the cipher in decryption mode.

STEP-7: Decrypt the ciphertext using the same secret key.

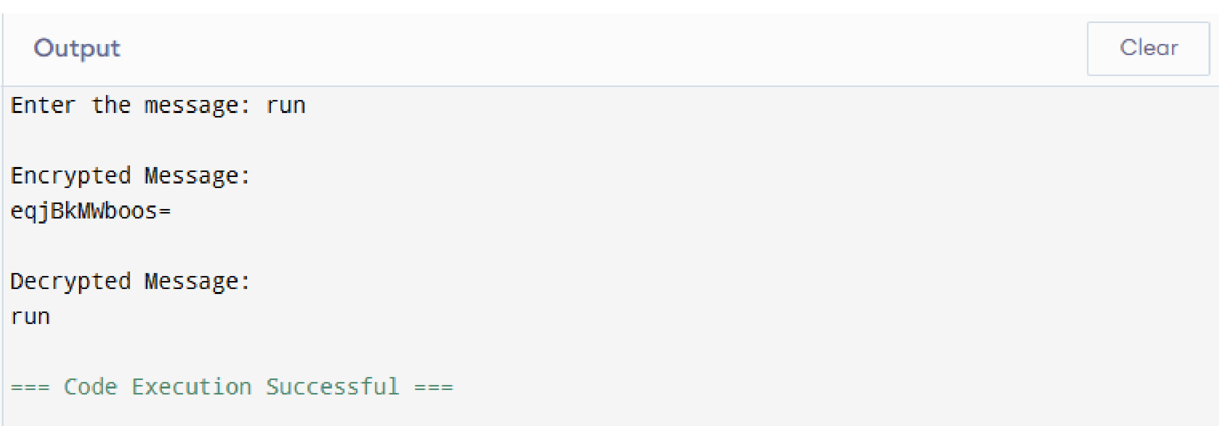
STEP-8: Display the original plaintext.

CODE

```
import javax.crypto.Cipher;

import javax.crypto.KeyGenerator;

import javax.crypto.SecretKey;

import java.util.Base64;

import java.util.Scanner;


public class TripleDES {


    public static void main(String[] args) {
```

```
try {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter the message: ");
    String message = sc.nextLine();

    // Generate Triple DES Key
    KeyGenerator keyGenerator = KeyGenerator.getInstance("DESede");
    keyGenerator.init(168);

    SecretKey secretKey = keyGenerator.generateKey();

    // Encryption
    Cipher cipher = Cipher.getInstance("DESede");

    cipher.init(Cipher.ENCRYPT_MODE, secretKey);

    byte[] encryptedBytes = cipher.doFinal(message.getBytes());

    String encryptedText =
            Base64.getEncoder().encodeToString(encryptedBytes);

    System.out.println("\nEncrypted Message:");
    System.out.println(encryptedText);
```

```
            // Decryption
            cipher.init(Cipher.DECRYPT_MODE, secretKey);

            byte[] decryptedBytes = cipher.doFinal(encryptedBytes);

            String decryptedText = new String(decryptedBytes);

            System.out.println("\nDecrypted Message:");
            System.out.println(decryptedText);

        } catch (Exception e) {

            e.printStackTrace();

        }

    }

}
```



```
Output                                                    Clear
Enter the message: run

Encrypted Message:
eqjBkMWboos=

Decrypted Message:
run

=== Code Execution Successful ===
```

RESULT

Thus, the Triple Data Encryption Standard (3DES) encryption and decryption algorithm was implemented successfully using Java.